

Program : Diploma in Wood and Paper Technology	
Course Code : 4128	Course Title: Paper Technology Lab
Semester : 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To familiarize the students with testing of various properties of paper
- To introduce the students with the preparation of paper board

Course Prerequisites:

Topic	Course code	Course name	Semester
Paper properties		Paper technology	4
Pulping		Pulp technology	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Test basic properties of paper like Basis weight, thickness	9	Applying
CO2	Test strength properties of paper like Surface strength, Bursting & Tensile	15	Applying
CO3	Test physical properties of paper like smoothness, Softness & Porosity	12	Applying
CO4	Test optical properties of Brightness and Opacity of paper	6	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2				3			
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Test basic properties of paper like Basis weight, thickness		
M1.01	Preparation of paper& paperboards from waste paper	3	Applying
M1.02	Test basis weight, caliper, thickness, density and bulk	6	Applying
CO2	Test strength properties of paper like Surface strength, Bursting & Tensile		
M2.01	Bursting strength and burst index	2	Applying
M2.02	Tearing strength and tear index	1	Applying
M2.03	Tensile strength and tensile index, stretch index	3	Applying
M2.04	Surface strength of paper(Wax pick)	3	Applying
M2.05	Folding endurance, stiffness	3	Applying
M2.06	Sheet evaluation for strength properties	3	Applying
	Lab Exam – I	1	
CO3	Test physical properties of paper like smoothness, Softness & Porosity		
M3.01	Test Smoothness, softness and porosity Water absorbency – Cobb’s method, Degree of sizing ash content, moisture content	12	Applying
CO4	Test optical properties of Brightness and Opacity of paper		
M 4.01	Brightness, yellowness and Opacity of paper	6	Applying
	Lab Exam – II	2	

Text / Reference:

T/R	Book Title/Author
T1	K.W.Britt, Hand book of Pulp and paper technology, CBS Publishers, New Delhi
R1	Tappi Test Methods for pulp testing, T 1-200 series
R2	Tappi test Methods for paper Testing, T 400-500

Online Resources:

Sl.No	Website Link
1	https://www.bellmer.com/paper-technology/
2	https://www.pulpandpaper-technology.com/
3	https://www.youtube.com/watch?v=I3SodZtes-c